The AI Technology Stack

and Why It Matters for AI Policy and Governance

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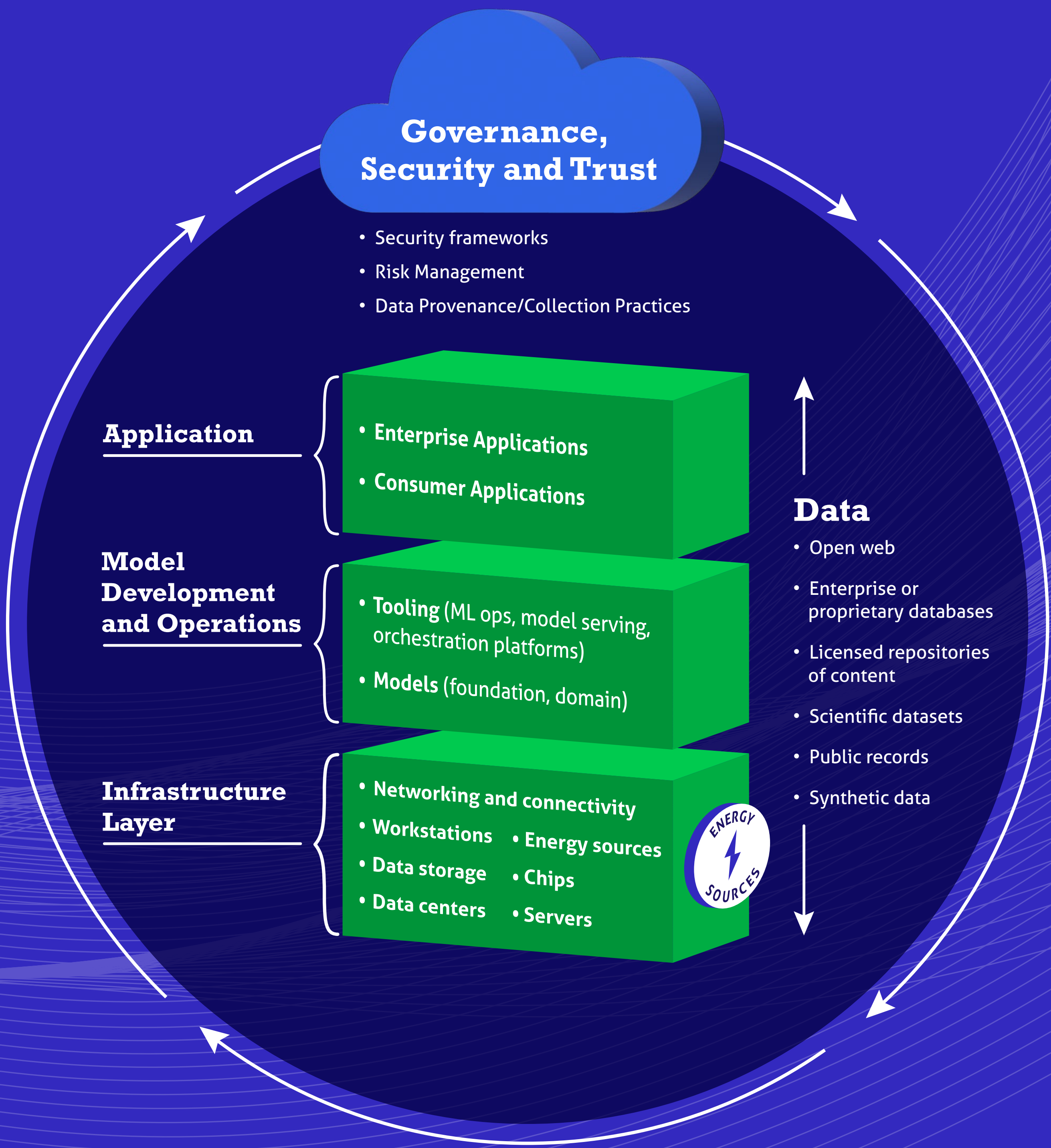
What is the AI Technology Stack?

When we think about artificial intelligence (AI), we often focus on chatbots that generate text, tools that create images, or companies that develop large language models, like GPT or Claude. But sitting alongside these tools and models is an ecosystem of upstream and downstream infrastructure and technologies—known as the AI technology stack—that makes AI development and deployment possible.

ITI represents companies from across the AI technology stack.[1] This explainer outlines how we think about the AI technology stack, including why it is important to policymaking. We break it down into five layers—infrastructure, data, model development and operations, application, and a cross layer of governance—to illustrate who is involved at each stage and why it matters for policymakers.[2] Many providers participate across multiple layers of the tech stack, with some such as hyperscalers offering full stack solutions that provide remote access to the hardware, data, and applications. For clarity, we classify capabilities by function: cloud infrastructure sits in the Infrastructure layer; managed MLOps tools in Model Development and Operations; foundation-model APIs in the Model Development and Operations layer, and vendor-built end-user services in Applications.

A shared understanding of the entire AI technology stack will help develop effective and tailored policies, address risks across various layers, and ensure that policymaking keeps pace with technological change.

Detailed Overview of the AI Technology Stack



1 Infrastructure Layer

The base layer of the AI technology stack is comprised of hardware, physical, and digital infrastructure that powers AI systems, particularly foundation models.

Chips: Semiconductors, or chips, are the foundational hardware layer of the AI technology stack. They provide compute power, memory, networking, and storage for the vast amounts of data and information required to build AI and perform AI tasks (whether locally or remotely) in data centers and at the edge. These chips include specialized logic chips called GPUs (graphics processing units), CPUs (central processing units), NPUs (neural processing units), TPUs (tensor processing units), and other custom AI accelerator chips that enable the development, training, and inferencing of resource-intensive complex AI models and applications. Chips used for AI also include networking, memory and storage chips like DRAM (dynamic random-access memory), HBM (high bandwidth memory), and NAND.

The semiconductor industry includes a variety of actors, including fabless logic chip designers, hyperscalers that design their own ASICs (Application-Specific Integrated Circuit) logic chips for their internal or on device use cases, fabless networking chip designers, and integrated device manufacturers (IDMs). All fabless companies outsource production to foundries that specialize in large-scale chip manufacturing, while IDMs design and manufacture their own chips in-house.

The AI chip supply chain is also dependent on material engineering companies involved in wafer production (semiconductor equipment manufacturers).

Materials engineering innovations are enabling chipmaker roadmaps in logic, DRAM, high-bandwidth memory (HBM) and advanced packaging, all of which are critical to improving chip performance-per-watt.

Compute power required to support the growth of AI is driving energy consumption to new levels and, as such, traditional 2D scaling of devices is no longer sufficient. To deliver improved performance and lower power, all devices are becoming 3D, which requires new materials and technologies.

Furthermore, the AI supply chain also requires companies specializing in hyper-polysilicon production, electronic design automation (EDA), as well as lithography, assembly, testing, and packaging. AI systems are not only increasingly relying on 3D stacking, but on advanced packaging including heterogeneous integration of logic, memory, and other components to meet the growing AI demands for speed, efficiency, and scalability.

Significant investments in R&D pertaining to material engineering solutions, chip design, and large-scale chip manufacturing have led to the development of smaller, faster, and more energy-efficient chips that have been essential to AI innovation.

Servers: GPUs, CPUs, custom ASICs, and memory are integrated into high-performance servers optimized for AI and machine learning workloads. AI servers support AI workloads from training to deployment and are often deployed in data centers, whether hyperscale, on-premises, or at the edge. At the highest end of the high performance computing spectrum are supercomputers, which aggregate thousands of processors and specialized chips into tightly coupled systems capable of training the largest AI models and running advanced simulations at unprecedented speed.

Workstations: High performance workstations integrate CPUs, GPUs, and memory into a single system that enables researchers, engineers, and AI developers to build, train, and test models locally. Unlike servers and data centers designed for large scale distributed training, workstations provide dedicated, readily accessible compute power that accelerates experimentation and iteration. They are critical for prototyping AI workflows, fine-tuning models, and ensuring access to compute in environments where cloud resources may be unavailable or impractical.

Data storage: Large-scale data storage—including on HDDs and SSDs—is necessary to handle the vast amounts of data required for data ingestion, training AI models, and inference processes. This includes setting up data lakes or high-capacity, fast-access storage systems.

Networking and connectivity: AI workloads depend on high-speed network fabric that links compute and storage within racks, across the data-center campus, and over long-haul fiber between facilities.[3] This fabric is built from fiber optic cables, connectivity products, and network equipment (e.g. switches, routers, and optical systems) engineered for low latency and high throughput. Optical fiber, cable manufacturers, and network equipment providers are continuously innovating to support the massive bandwidth demands of generative AI. Mobile networks are equally critical. Today, AI is increasingly embedded within the IoT ecosystem, where decisions need to happen instantly and autonomously. Consumers, industry, and first-responders will also demand access to AI-enabled applications anytime, anywhere. Robust mobile networks make this possible, delivering the speed, coverage, and responsiveness required. This mobility enables AI to operate in real time, anywhere it is needed. Additionally, AI-driven Radio Access Networks (AI-RAN) are emerging as critical enablers by using AI to dynamically manage spectrum, optimize traffic flows, and improve network efficiency.

Data centers: AI data centers—facilities that host servers, chips, and other hardware that power AI workloads—are at the center of the physical infrastructure layer.[4] Cloud service providers, large social media companies, and other large digital data holders are often referred to as hyperscalers. The large-scale facilities they operate are called hyperscale data centers, which are designed to handle large amounts of data and the computing tasks of multiple organizations. Hyperscale data centers are the most common data center model for running AI workloads as they support high-density rack configurations, advanced cooling, and ultra-fast interconnects, as well as the on-demand scalability required for AI training utilizing thousands of GPUs. Data center companies, including hyperscalers, also offer global data center footprints, edge locations, and regional availability zones that benefit AI use cases that need data locality, regulatory compliance, or global inference performance. Hyperscale companies either build, own and operate their own physical data centers, or lease entire facilities or large portions of space from third-party data center operators. In both cases, the hyperscaler manages and maintains its own servers and IT equipment, while the third-party provides and maintains the underlying infrastructure. Companies that develop AI are increasingly investing in all models of AI data centers in the United States and across the globe and often work with local governments and utilities to secure suitable land, access power grids, and meet power requirements. In addition, AI data center developers are increasingly using sophisticated modeling software to improve the design and layout of data centers to minimize their energy needs.

Energy sources: While data centers consider many factors during the site selection process, an important consideration in the infrastructure layer is access to reliable sources of electricity. Data centers either "co-locate" with existing power plants, connect to the grid, or build their own on-site power generation off-grid in cases where grid interconnection is too burdensome. Building behind-the-meter generation, however, requires significant up-front capital investment, faces challenges from ongoing long-term supply chain constraints, and triggers additional permitting processes. As a result, data centers often connect to the grid, if possible. Because hyperscale data centers must meet intense computing processing and storage demands of training and deploying AI models in the cloud, they require vast amounts of energy, high-speed connectivity, and advanced cooling systems.[5] Many hyperscalers are investing directly in new power generation capacity to ensure sufficient electricity is available to support the primary data center infrastructure supporting AI workloads.

2 The Data Layer

Data: The quality, representativeness, and reliability of AI systems largely relies on the data they are trained on and have access to while operating. Through data, AI systems learn language, recognize patterns, and make predictions. This process begins with data curation which involves the selection, preparation, and management of datasets to ensure their quality and relevance to a certain AI task. Understanding the provenance of this data—its origins, history, legal basis, and lineage—is crucial to responsible AI development.

The data sources vary and can include the open web, enterprise or proprietary databases, licensed repositories of content, scientific datasets, public records, and artificially generated synthetic data. The actors operating in this space include companies that generate specialized datasets, public sector agencies, researchers, data brokers, and the company building or deploying the AI system. Some companies supply raw data, while others offer specialized services to annotate, validate, and evaluate data for AI models. Data is also generated downstream and can be fed back in during the model training phase, creating a continuous feedback loop.

3 The Model Development and Operations Layer

This layer of the AI technology stack includes tooling that enables model training and optimization, as well as AI models. Importantly, AI model training draws on traceable data pipelines created as a part of the data layer (referenced above).

Tooling (Platforms/MLOps)

MLOps platforms: These platforms offer a set of tools that helps teams build, test, and manage machine learning models more efficiently.[6]

Model serving: It is the process of hosting machine learning models on cloud infrastructure and making their functions available via an API so users can incorporate them into their own applications.[7]

Resource sharing software: Tools that allow organizations to pool and allocate high demand resources, like GPUs, across workstations and end user devices. They reduce cost, expand access to compute, and support in scaling secure, efficient AI development.

Orchestration platforms: These platforms automate AI workflows, track progress toward task completion, manage resources, monitor and document data flows and memory, and handle errors.[8]

AI software frameworks simplify the process of creating and training AI models. Instead of starting from scratch, AI developers often use these tools to help accelerate and streamline AI tasks.[9]

Enterprise/inference deployments: Once a model is trained and validated, it can be used for fine-tuning and inference—the process of making predictions or classifications with new, unseen data. While training is computationally intensive and typically performed in powerful data centers, inference often happens in real time and can occur anywhere, from cloud servers to edge devices in remote locations. Enterprise customers leverage AI compute—both cloud-based and on premises—for fine-tuning and inference in a wide range of industry applications (discussed further in the Applications layer).

API Management: Application Programming Interface, or API, is a set of rules or protocols that enable communication between various components of the software application ecosystem.[10] API management includes tools and services that enable building, analyzing, operating, and scaling APIs in secure environments.[11] Public-facing API gateways and developer API management are discussed in the Applications section.

Models (Artifacts)

Foundation models: Foundation models are a central element of the model development layer. Foundation models are AI models that can be used as the foundation for many applications that are trained on diverse datasets, and which can be adapted for a variety of downstream tasks.[12] Examples that have been widely discussed in public discourse include GPT, Gemini, LLaMA, and Claude.[13] Once developed, foundation models can be fine-tuned or integrated into applications across industries. The development of these models often involves large teams, cutting-edge research, and significant financial investment.

Domain models: Domain-specific models may also be at the model development layer of the AI technology stack. These models are trained on specific data to perform a particular task or set of tasks and are applicable to specific domains. For example, MedPaLM is a model that is designed for the medical domain, trained on medical and biomedical text and code and FinBERT is designed for the financial domain, trained to analyze sentiment in financial documents.[14]

4 The Application Layer

The application layer is the part of the AI technology stack that most people interact with during daily life. This is the layer where AI systems are integrated into the tools we use at home, at work, and across industries, and where end-users interact with their outputs. Ultimately, this layer is crucial for generating the social and economic value that will convey the benefits of AI.

AI applications are delivered in three ways: via private or public cloud-hosted services, edge/on-device/on-premises deployments (e.g. AI PCs with integrated NPUs for local inference, phones, vehicles, AR/VR, industrial IoT, smart-home devices) and hybrid systems that split work between device and cloud. On-device inference is used when latency, privacy, intermittent connectivity, or cost favor keeping computation more local; the cloud is often used for heavier workloads.

Another key deployment model is AI as an extension to Software-as-a-Service (SaaS) products. AI-enabled SaaS allows for smarter and more dynamic solutions, moving from static software to software that learns, adapts, automates tasks, and offers more tailored value to users.

This model is increasingly common in SaaS, with AI-powered SaaS allowing usage-based pricing, improved workflow automation, better scalability, and personalized user experience.

Perhaps the most well-known examples of AI applications include OpenAI's ChatGPT or Anthropic's Claude, popular generative AI chatbots used by millions of people daily. Although ChatGPT and Claude are part of the applications layer, the foundation models that underpin them —GPT and Claude—fit into the model development layer of the AI technology stack. Turning a foundation model into an AI application—which may be done by either the model creator or another party with appropriate license to use that model—involves building a user interface, integrating an API, and adding other features such as plugins that allow the model to interact with external systems or perform specialized tasks.

Besides generative AI chatbots, many of today's software tools are AI-powered. In practice, they integrate AI features into traditional applications to boost productivity and streamline processes. The end use of AI-powered tools is an important component of the application layer. Personal or consumer-facing use cases include word processors that suggest phrasing, email platforms prioritizing messages, and video conferencing tools offering live transcription and translation. Enterprise use cases range from real-time analysis of medical images and patient monitoring in healthcare, to customer support, to predictive maintenance, and process optimization at the factory floor in manufacturing, to fraud detection and credit scoring in finance, to traffic management and autonomous vehicles in transportation.

In addition, a growing number of devices run features directly on the edge. For example, smartphones now offer faster photo processing, smart assistants, real-time translation, personalized user experience, and other AI-powered features.[15] Smart glasses enable hands-free interaction and information retrieval.[16] Smart home devices create new solutions to improve convenience and home security. For instance, AI virtual assistants can perform a variety of tasks with voice instructions, AI alarm systems can activate on their own, and AI home cameras can analyze events in real time.[17] Vehicles use AI for driver assistance and navigation. Companies are also integrating AI into industry-specific products. For instance, AI-powered medical tools can assist with surgeries or radiation therapy.[18,19] In manufacturing and logistics, AI-powered robots help automate assembly lines and accelerate product delivery.[20,21]

The Governance, Security, and Trust Overlay

Governance is a cross-cutting function that permeates every layer of the AI technology stack —from infrastructure to applications. Governance includes tools and frameworks for data collection and use, including specific to provenance, compliance, security-by-design, auditability, transparency, and traceability that help measure and manage risks. Governance approaches will differ based on the part of the stack that an organization is targeting, and decisions oftentimes flow down. Security is a key part of governance. As organizations adopt AI, safeguarding models, software, and data against AI-enabled threats—such as model manipulation, data breaches, and adversarial inputs—is a continuous practice. At the same time, the responsible use of AI can augment cyber defense by enhancing threat detection and accelerating response times, thus strengthening resilience.

Why Does the AI Technology Stack Matter for Policymaking?

Understanding the AI technology stack is integral to policymaking, especially as recent policy conversations have largely focused on chips and foundation models. Yet, effective and proportionate policymaking requires an understanding of the entire stack and the market participants and dynamics, recognizing that each layer may raise separate questions, present differing challenges for consumers, and in turn, benefit from different policy approaches. As policymakers respond to these questions and seek to promote AI innovation, appropriate and effective policy must consider the ways that different elements of the stack present risks and the capabilities of each actor to control and mitigate those risks. A common understanding of each layer will help ensure that policymakers, industry, and consumers are aligned and work from a shared foundation. Taken together, this can drive tailored policymaking across a range of areas, including trade, national security, and consumer protection.

Illustrative Layer-Specific Policy Tools

Infrastructure layer policies may include incentives to support domestic chip manufacturing and supply chain onshoring, grid capacity planning, renewable energy, open standards to promote interoperable infrastructure, and new energy efficiency standards for data centers. In the context of trade and security policies, it is critical that policymakers recognize that tariffs, export controls, and supply chain requirements of chips and equipment necessary for the AI stack can raise costs, slow innovation, and fragment global supply chains critical for AI development and deployment—especially where key products such as GPUs or infrastructure software may not be interoperable with other layers of the stack.

At the data layer, policy activities might include adopting a comprehensive federal privacy law, establishing data governance frameworks, including considering the role of data provenance tracking, defining how data is collected, stored, and used, as well as addressing risks related to data privacy and security.

At the model development and operations layer, to advance model-level risk management, policies may focus on pre-deployment capability evaluations, increased investments in measurement science, and the clear allocation of responsibilities among the various actors involved in developing and deploying these models.[22]

At the application layer, policymakers may focus on reviewing existing sector-specific rules and requirements. After such a review, policy discussions may focus on whether new rules and requirements are necessary to facilitate responsible deployment. In addition, policymakers may need to distinguish between enterprise- and consumer-facing applications as deployment in different contexts raises different considerations.

In some instances, **policymaking approaches may implicate multiple layers of the AI technology stack.** For example, a regulatory approach that erects minimum standards might implicate both the model development and application layer of the stack to appropriately distribute responsibility. In the national security context, conversations are increasingly considering different layers of the stack. Evaluating risks and opportunities at each layer enables policymakers to better develop policy approaches that address U.S. national security risks while promoting the U.S. global leadership and competitiveness that is part and parcel of advancing U.S. national security.

Anchoring policymaking in the context of the AI technology stack ensures that policy frameworks and legislative approaches are appropriately targeted to risks and remain flexible to better keep pace with technological change.

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